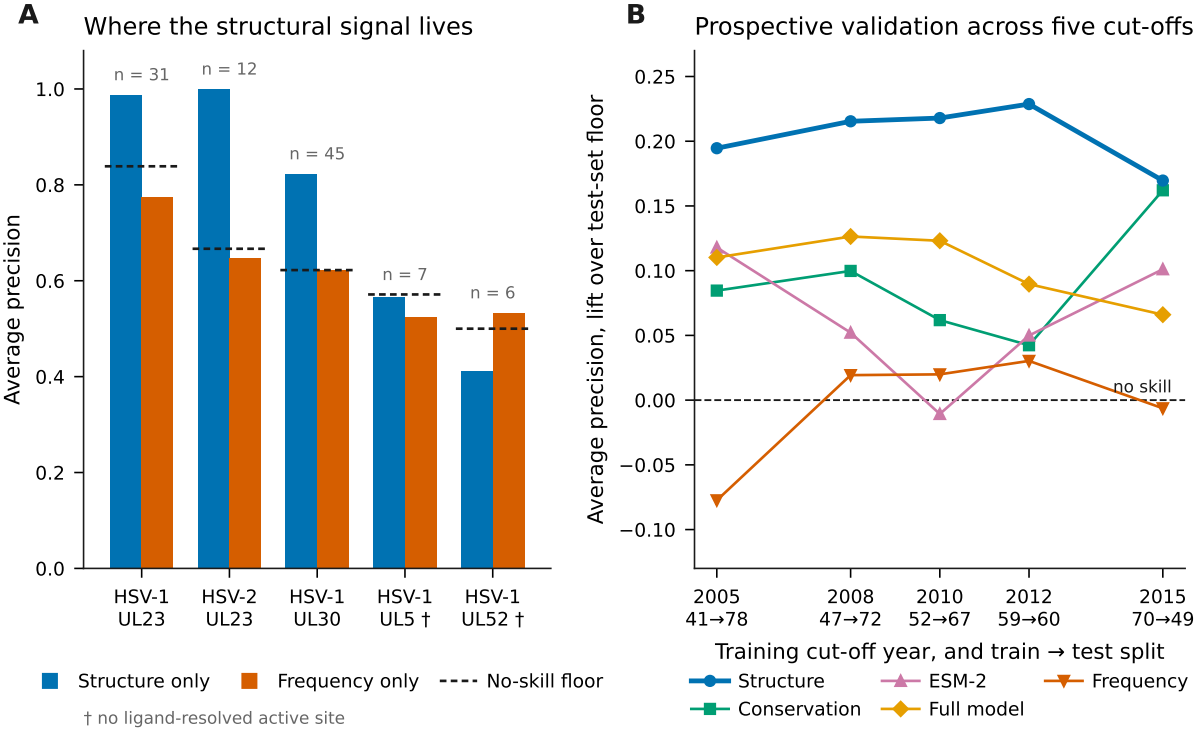
A
Where the structural signal lives
Average precision
n = 31
n = 12
n = 45
n = 7
n = 6
HSV-1 UL23
HSV-2 UL23
HSV-1 UL30
HSV-1 UL5 †
HSV-1 UL52 †
Structure only
Frequency only
No-skill floor
† no ligand-resolved active site
B
Prospective validation across five cut-offs
Average precision, lift over test-set floor
no skill
2005 41→78
2008 47→72
2010 52→67
2012 59→60
2015 70→49
Training cut-off year, and train → test split
Structure
ESM-2
Frequency
Conservation
Full model